

Experimental Evaluation

Text-Driven Face Editing using StyleCLIP-NADA

The objective of this experimental study is to investigate the influence of different training strategies on the quality of text-driven face editing using StyleCLIP-NADA.

Experimental Setup

In all experiments, we used a pre-trained StyleGAN2-Ada model trained on the FFHQ dataset. Text-based control of image generation was performed using CLIP ViT-B/32. The model was trained with the Directional Loss, which encourages the direction of image changes to align with the corresponding text description in the CLIP feature space. The Adam optimizer was used for training.

Experiment 1. Selection of Trainable Generator Components: Mapper Training

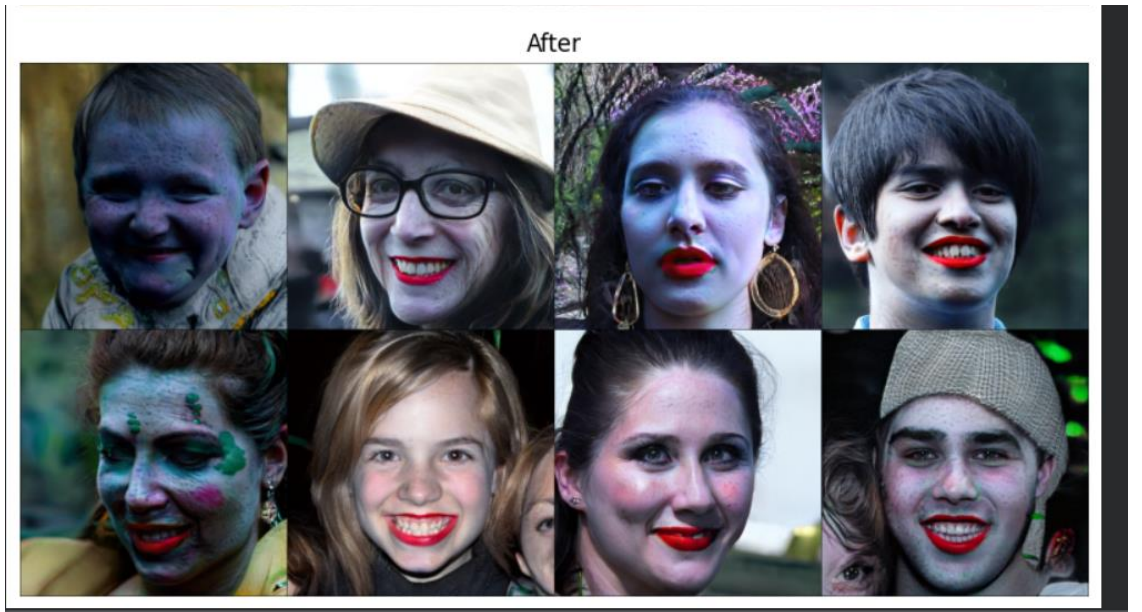
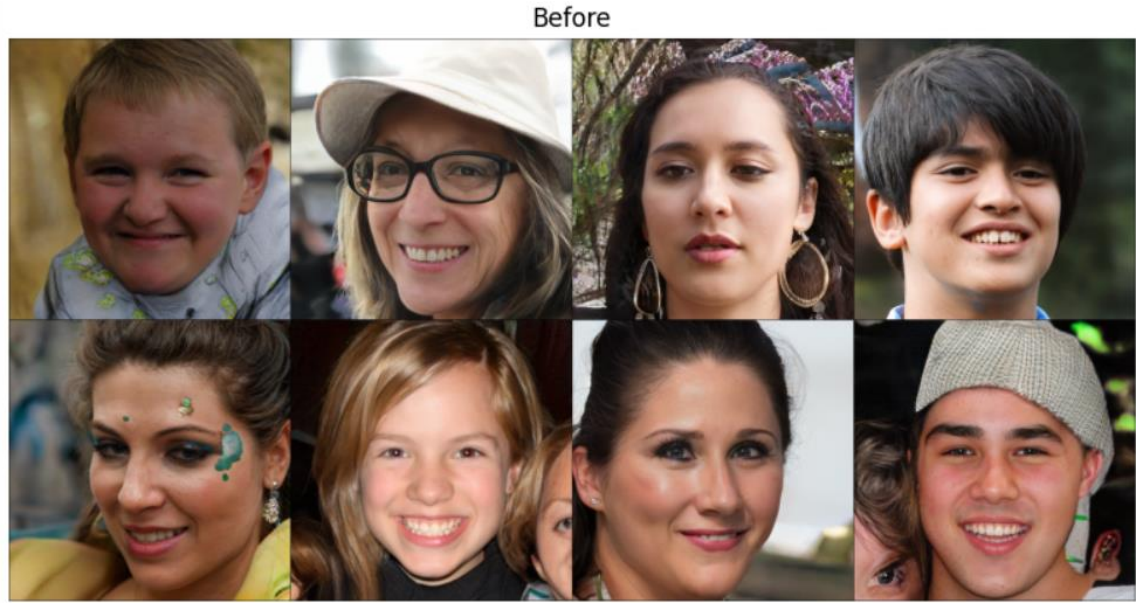
Objective. To investigate whether text-based image editing can be achieved by training only the Mapper network, while keeping the StyleGAN2 generator parameters fixed.

The learning rate was set to $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0008. The latent vector was constrained to satisfy ($\|w\| < 2.0$).

The following prompts were used during training:

Source: “a photo of a person”

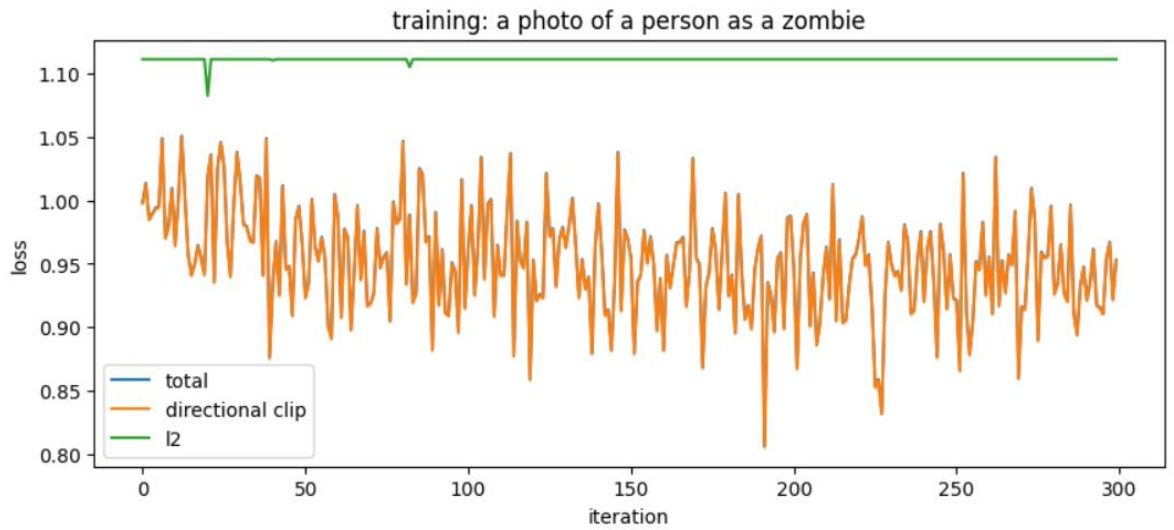
Target: “a photo of a person as a zombie”



The following observations can be made based on the experimental results. The generator preserved the overall facial structure, while the color characteristics of the images changed. The skin acquired a bluish tint, shadows around the eyes became more pronounced, and the faces appeared paler. However, pronounced stylization toward the target “zombie” appearance was not achieved. The changes primarily affected the color characteristics of the images, whereas the facial shape and most of the textures remained largely unchanged.

In addition, a general effect of increased lip redness can be observed across nearly all images. This effect does not correspond to the target style and suggests

that the model converged to a local minimum of the loss function.



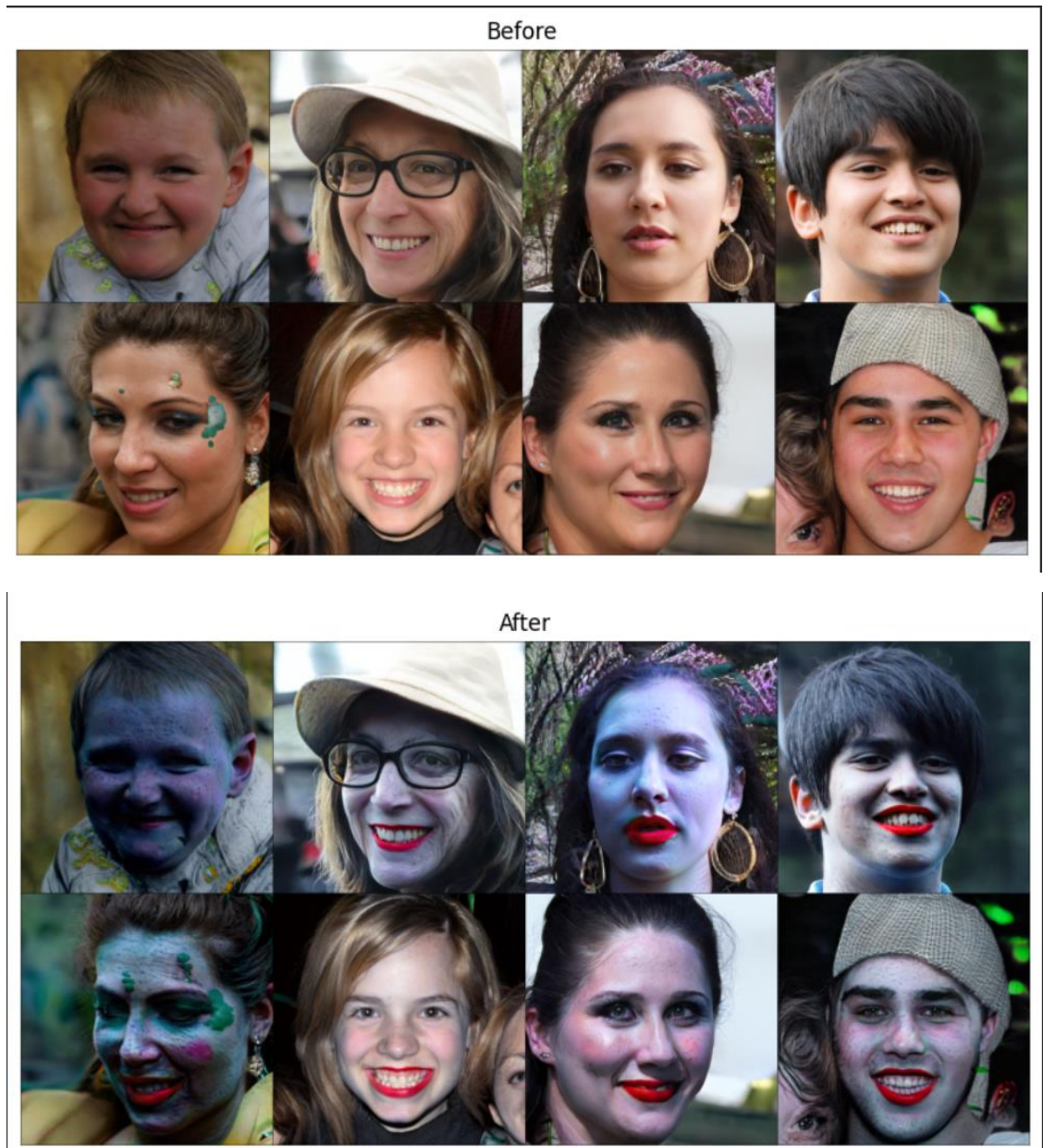
The figure shows the evolution of the Total Loss, Directional CLIP Loss, and L2 regularization terms during Mapper training for transforming a human image toward the “zombie” style in the first experiment. During training, the Directional CLIP Loss gradually decreased from approximately 1.0 to approximately 0.93. The overall trend indicates a gradual improvement in the alignment between the image changes and the given text description.

Conclusion. Training only the Mapper network allows the latent representation of an image to be modified. However, the capabilities of this approach are limited. Although some features of the target style emerge, the resulting stylization remains insufficient.

Experiment 2. Training the Mapper and Mapping Network

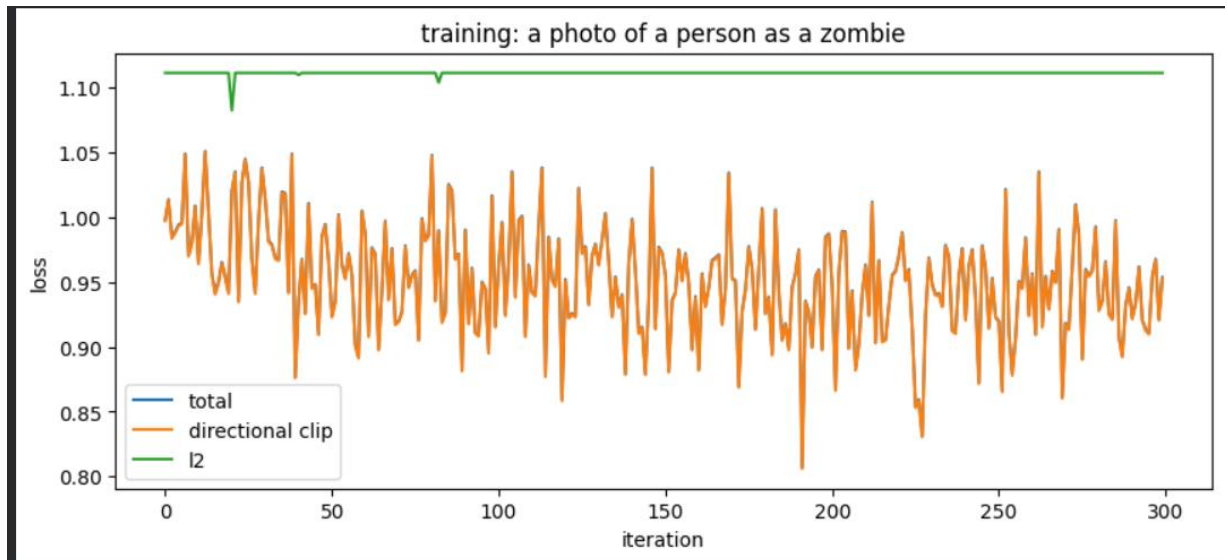
Objective. To investigate the effect of jointly fine-tuning the Mapping Network and Mapper on the quality of text-guided style transfer.

The learning rate was set to $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0008. The latent vector was constrained to satisfy ($\|w\| < 2.0$).



Compared to the first experiment, the changes became more pronounced. The skin acquired a stronger blue tint, the darkening around the eyes and facial areas became more noticeable, and the images appeared darker overall, while the facial structure remained almost completely preserved.

However, the style transfer remained limited. Instead of introducing characteristic zombie features, such as skin deformation, pronounced damage, or changes in facial texture, the model primarily altered the color palette of the images. The general effect of red lips appearing on nearly all faces also persisted, which cannot be considered a desirable stylization result.



After unfreezing the Mapping Network, the Directional CLIP Loss decreased slightly faster compared to the first experiment. The fluctuations in the loss value also became somewhat more pronounced, which can be attributed to the increased number of trainable parameters.

Despite this, the training remained stable, and the L2 regularization continued to effectively constrain changes in the latent space. These results indicate that the Mapper and Mapping Network were successfully optimized jointly.

Conclusion. Fine-tuning the Mapping Network produces stronger image changes compared to training the Mapper alone. However, the changes still primarily affect the color characteristics of the images rather than their textural features.

Experiment 3. Joint Training of the Mapper and the Last Generator Block (synthesis.b1024)

Objective. To investigate the effect of unfreezing the last block of the StyleGAN2 generator (synthesis.b1024) on the quality of text-based image editing. It was hypothesized that fine-tuning the final generator block would allow the output image to be better adapted to the text description while preserving the overall facial structure.

The learning rate was set to $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0008. The latent vector was constrained to satisfy ($\|w\| < 2.0$).

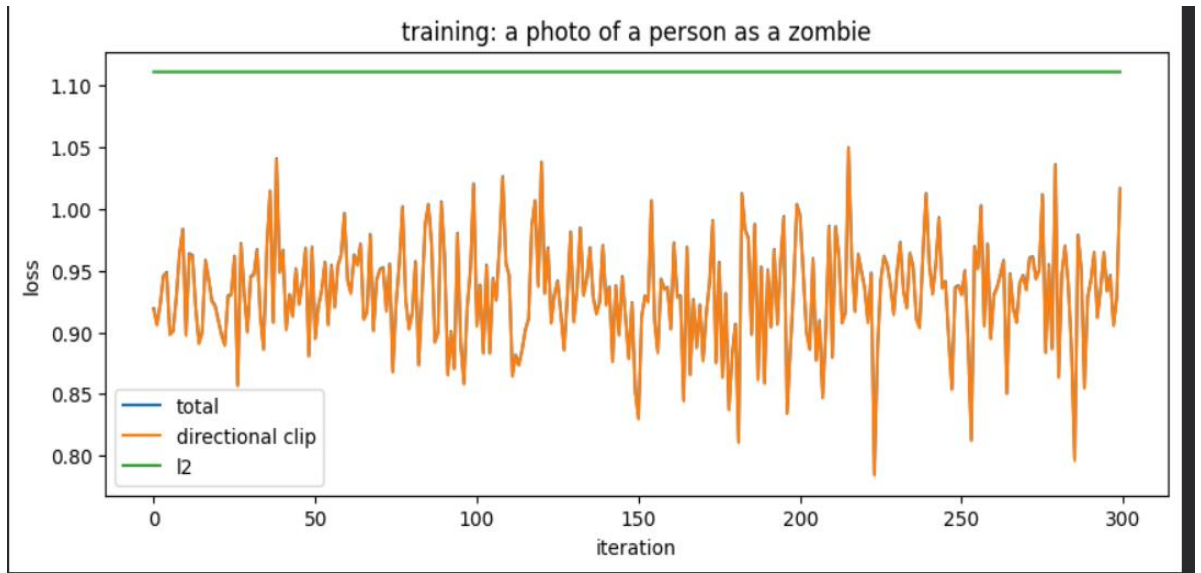
Before



After



The changes became significantly more pronounced. The images exhibit several noticeable effects: the skin acquired a distinct bluish-green tint, the faces became paler, the lips became more vividly red, and the dark areas around the eyes became more pronounced. Overall, the color balance shifted closer to the target “zombie” appearance.



During training, the Directional Loss fluctuated approximately within the range of 0.8–1.05. The plot shows regular fluctuations; however, there are no signs of an increasing loss or divergence during training.

The L2 regularization term remained nearly unchanged, maintaining a stable value of approximately 1.1 and limiting the magnitude of the latent shift.

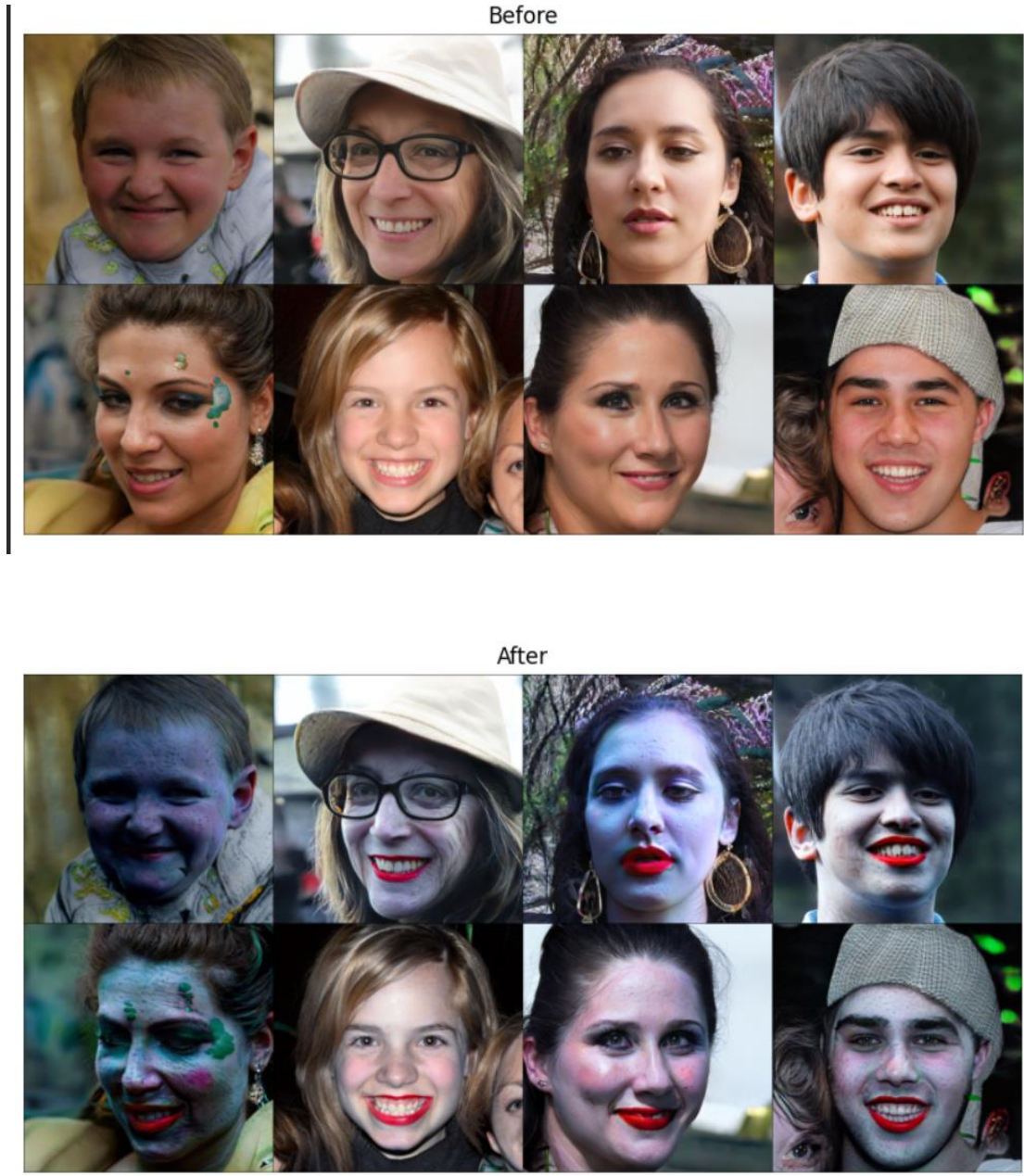
Despite the presence of local fluctuations, the optimization process remained stable, and the model successfully adapted to the target text description.

Conclusion. Unfreezing the last generator block substantially increased the influence of the text description on the generated output. Compared to training the Mapper alone, the resulting changes became more pronounced, particularly in the color characteristics of the images. At the same time, the facial structure was preserved, and severe geometric distortions were avoided.

Experiment 4. Joint Training of the Mapper and the synthesis.b512 and synthesis.b1024 Blocks

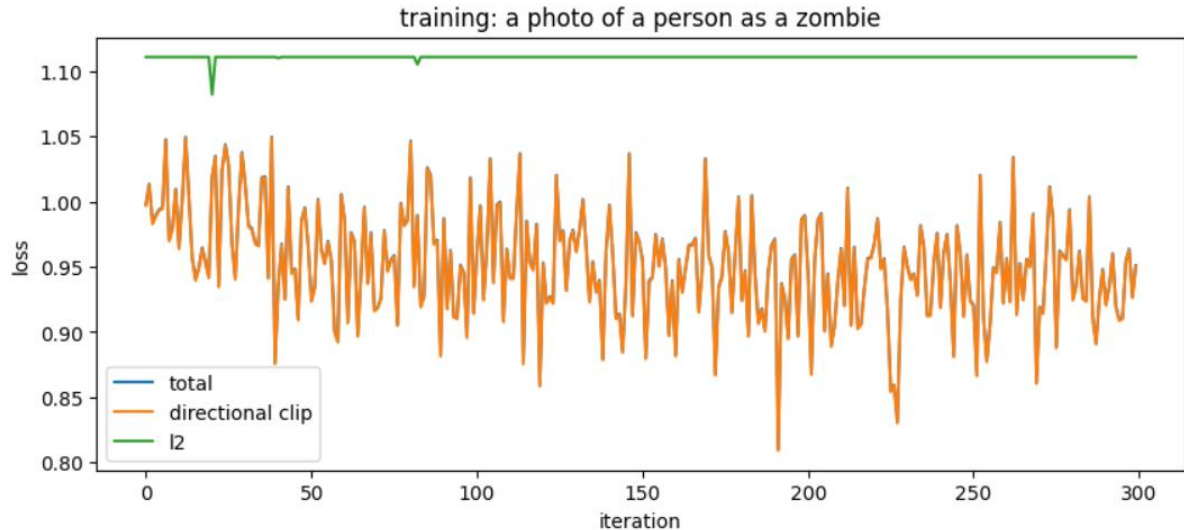
Objective. To investigate the effect of additionally unfreezing the penultimate generator block (synthesis.b512) on the quality of text-based image editing. It was hypothesized that training the two final generator blocks would enhance the transfer of the text description by enabling more flexible adaptation of high-level image features.

The learning rate was set to $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0008. The latent vector was constrained to satisfy ($\|w\| < 2.0$).



After training, the images exhibited characteristic features of the target style: the skin became bluish and pale, the overall color tone shifted toward cooler hues, the lips acquired a saturated red color, and darker areas appeared around the eyes. The original facial structure was preserved.

However, the visual results were very similar to those obtained in the previous experiments. Additionally unfreezing the synthesis.b512 block had little effect on the degree of alignment between the generated images and the text description.



The behavior of the Directional Loss was nearly identical to that observed in the previous experiment. The loss remained within a similar range, exhibiting stable fluctuations without any signs of deterioration in the training process.

The L2 regularization term also remained nearly constant, indicating that the magnitude of changes in the latent space remained well controlled.

The resulting plots show that additionally unfreezing the synthesis.b512 block had no noticeable effect on the optimization dynamics.

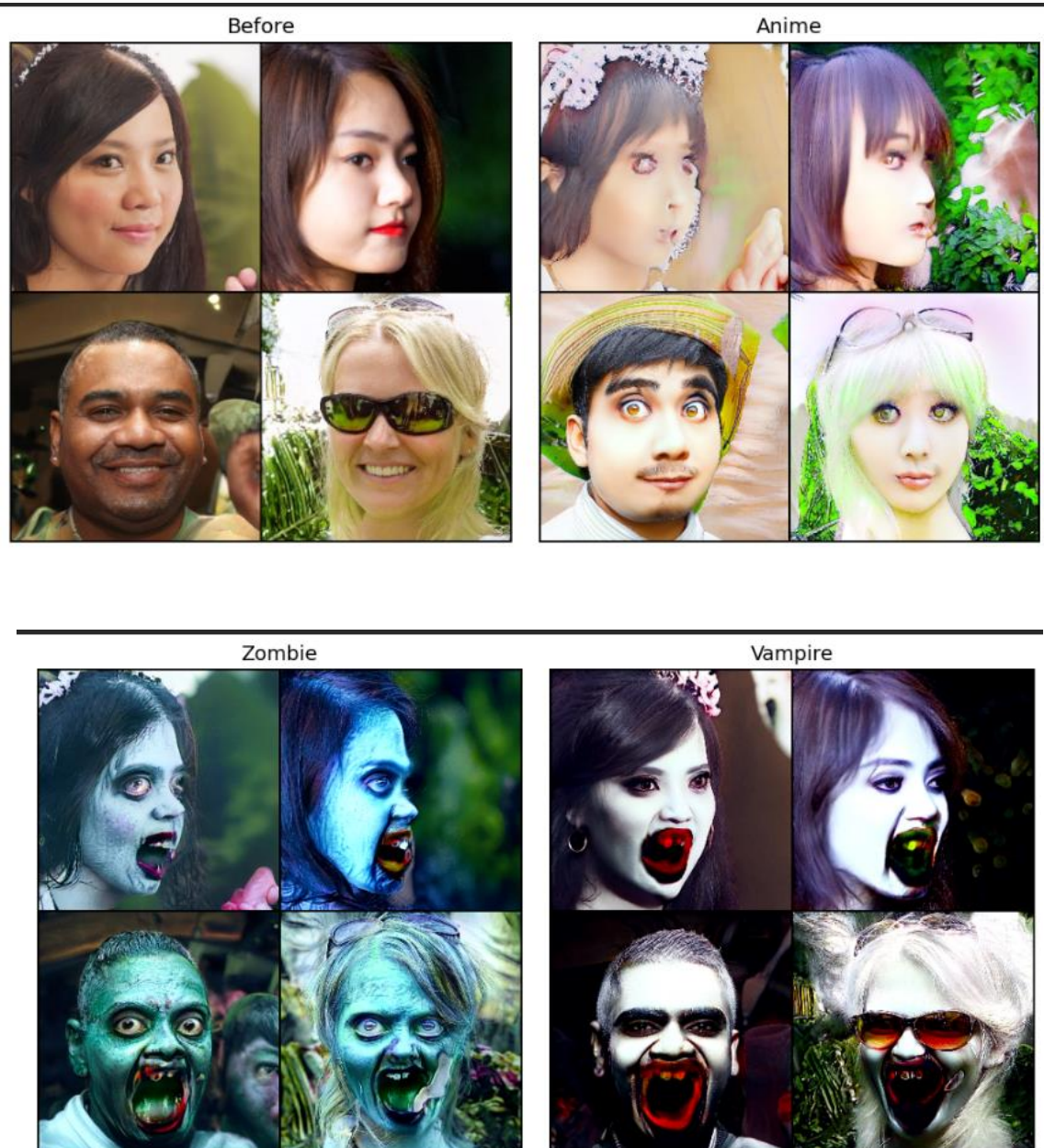
Conclusion. Unfreezing the two final generator blocks did not result in a noticeable improvement in stylization quality compared to training only the last synthesis.b1024 block. The images exhibited similar changes in color palette and skin texture; however, the degree of resemblance to the target zombie appearance did not increase substantially.

Experiment 5. Selection of Optimal Training Hyperparameters

Objective. To investigate the feasibility of training a single Mapper to simultaneously support multiple target styles.

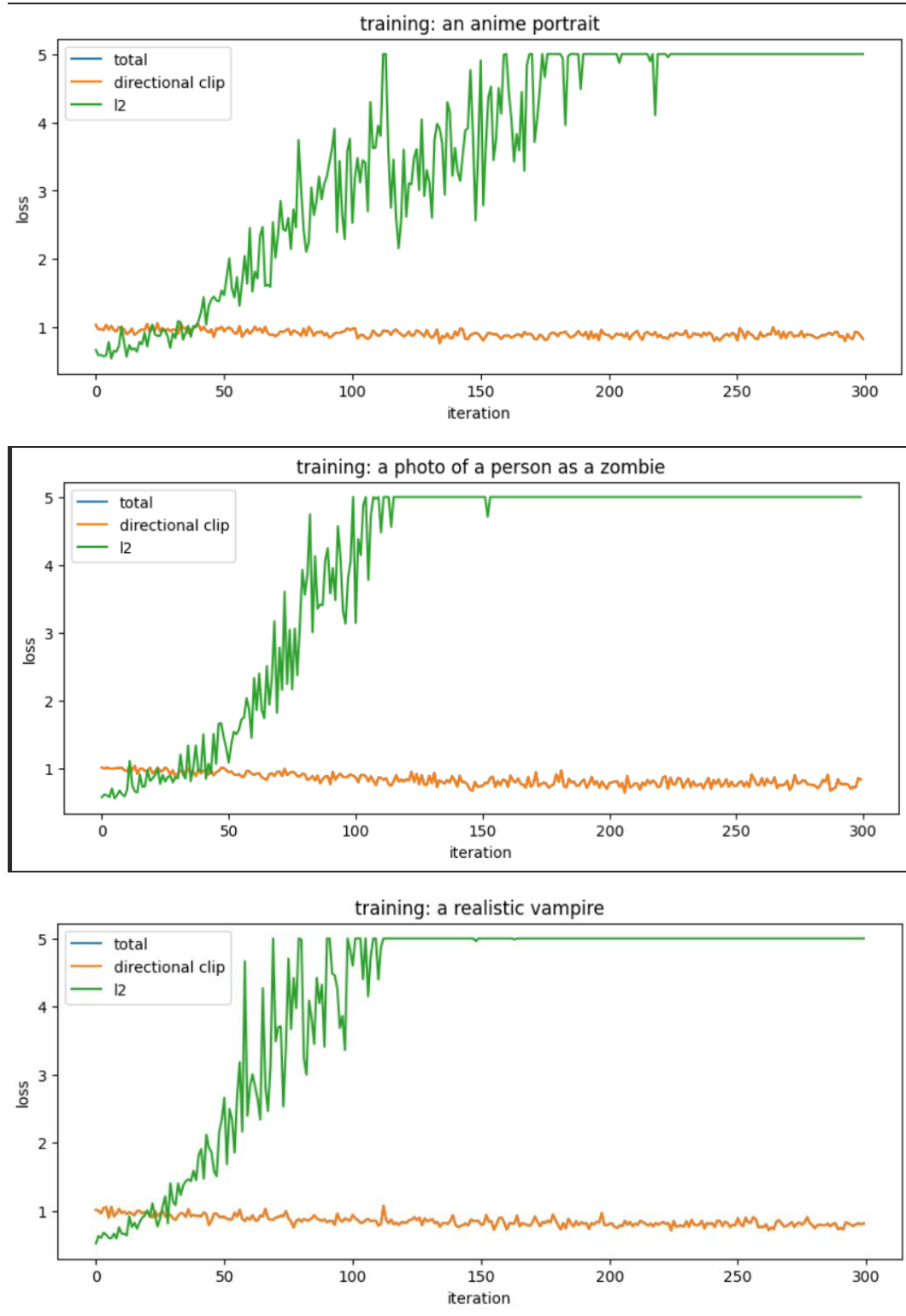
Unlike the previous experiments, training was conducted simultaneously for three different target styles: anime, zombie, and vampire. A separate Mapper was trained for each style using the same hyperparameter configuration, allowing the robustness of the selected parameters to be evaluated across different visual styles.

The learning rate was set to $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0006. The latent vector was constrained to satisfy ($\|w\| < 5.0$).



After training, the images exhibited the most pronounced characteristics of the corresponding target styles. In all cases, the main facial features and the identity

of the original subject were largely preserved despite some degree of deformation, although the extent of these changes varied depending on the target style.



Throughout training, the Directional CLIP Loss remained relatively stable and fluctuated within a narrow range without any pronounced increase or decrease. This

behavior indicates that the model gradually adapted to each of the target editing directions while maintaining a stable optimization process. The L2 regularization term increased gradually and reached its predefined limit after approximately 100–180 iterations, after which it remained nearly constant. This behavior was observed for all three target styles, indicating that the model utilized the maximum allowable latent-space shift to achieve the desired editing direction.

Conclusion. Despite the successful convergence of the training process and the stable behavior of the loss function across all three target styles, using a single shared Mapper did not achieve editing quality comparable to that obtained by training separate models for each style. These results suggest that, for the task of text-driven face editing, a highly specialized Mapper is more important than reducing the number of trainable models by combining multiple target styles into a single model.

Conclusion

In this work, a series of experiments was conducted to improve the quality of text-based image editing using the StyleCLIP-NADA method.

The initial experiments focused on varying the constraint on the magnitude of the latent representation shift, $\|w\|$. Both increasing and decreasing the maximum allowable change in the latent vector were investigated. However, no noticeable improvement in generation quality was achieved. Increasing the allowed magnitude of $\|w\|$ resulted in more aggressive image modifications, leading to pronounced facial deformations and reduced preservation of the subject’s identity. Conversely, a stricter constraint limited the model’s ability to transfer the characteristics of the target style, resulting in less pronounced stylization.

Experiments were also conducted to investigate the effect of varying the number of trainable blocks in the StyleGAN2 generator. In addition to training only the final synthesis.b1024 block, unfreezing the synthesis.b512 block was also explored. Despite the increased number of trainable parameters, no substantial improvement in the visual quality of the generated images was observed.

The most successful results were obtained by training only the final synthesis.b1024 block with a learning rate of $1e-4$, with 300 raining iterations and an L2 regularization coefficient of 0,0006. The latent vector was constrained to satisfy ($\|w\| < 5.0$).

Despite conducting a series of experiments involving changes to the model architecture and training hyperparameters, the quality of the resulting images remained below that reported in the original StyleCLIP-NADA paper. The most notable difference was the more pronounced alteration of facial geometry and the reduced preservation of the subject’s identity during target-style transfer.

One possible reason for this difference is the substantially shorter training duration compared to the original work. In this study, the Mapper was trained for only 300 iterations, whereas the original paper used a considerably longer optimization process accompanied by more extensive hyperparameter tuning. The limited number of iterations may reduce the Mapper’s ability to fully adapt to the editing direction in the latent space, causing some of the desired changes to be compensated for by modifications to the facial structure.

Another contributing factor is the relatively small training set. In the original implementation, training is performed using a substantially larger number of latent codes, allowing the Mapper to learn a more robust editing direction. In the present work, the training dataset was considerably smaller, which limits the model’s ability to generalize the learned direction of change in the latent space.

It should also be noted that the optimization was performed exclusively using the Directional Loss. This loss function is designed to improve the alignment between the generated image and the text description but does not explicitly constrain the preservation of the subject’s identity. Consequently, as the model seeks to achieve a higher degree of text-image alignment, it may modify not only the image style but also the facial geometry.

